Unsaid – A Mini Social Platform

Project Report

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Project Title: *Unsaid – A Mini Social Platform*

1. Introduction

In today's digital era, social media platforms have become a primary medium for communication and expression. While these platforms connect people globally, they often fail to provide a safe and judgment-free environment. Many users hesitate to express their thoughts due to fear of being judged, misunderstood, or exposed. This issue is particularly significant when individuals are emotionally vulnerable or experiencing loneliness.

Unsaid – A Mini Social Platform is designed to address this problem by enabling users to express their thoughts anonymously. The platform removes identity pressure, allowing users to share what they feel without revealing personal details. The goal is to create a space where people can speak freely, interact through comments, and feel emotionally supported.

From a technical perspective, this project demonstrates the implementation of a full-stack web application using modern backend technologies. It focuses on clean architecture, RESTful API design, database persistence, and frontend-backend integration. The system is intentionally designed to be simple yet scalable, making it suitable as an academic project while reflecting real-world software engineering practices.

2. Problem Statement

In modern society, many individuals lack a trusted space to express their thoughts openly. Fear of gossip, judgment, and social pressure prevents people from sharing what they truly feel. Existing social platforms are identity-driven, making honest expression difficult. To address this issue, *Unsaid* provides an anonymous platform where users can post freely without revealing their identity, encouraging open communication and emotional relief.

3. Objectives of the Project

The primary objectives of this project are:

- To design and implement an anonymous social posting platform
- To allow users to create posts and comments without identity exposure
- To provide a simple and engaging interface for interaction
- To demonstrate backend development using Spring Boot and REST APIs
- To store data persistently using a relational database
- To ensure clean, modular, and maintainable code architecture

4. Technology Stack Used

Backend Technologies

- Java (JDK 17+)
- Spring Boot
- Spring Data JPA
- Hibernate ORM
- RESTful APIs

Database

- H2 File-Based Database

Frontend

- HTML5
- CSS3
- JavaScript (Vanilla JS)

Tools & Environment

- Maven
- PowerShell
- Visual Studio Code
- Git (optional)

5. System Architecture

The system follows a layered architecture, which ensures separation of concerns and maintainability.

Layers

1. Controller Layer – Handles HTTP requests
2. Service Layer – Contains business logic
3. Repository Layer – Handles database operations
4. Entity Layer – Defines database models

Each layer communicates only with its adjacent layer, making the system modular and easy to extend.

6. Backend Design and Implementation

6.1 REST API Design

The backend exposes RESTful endpoints for managing posts and comments.

Post APIs

- GET /posts – Fetch all posts
- POST /posts – Create a new post
- POST /posts/{id}/like – Like a post

Comment APIs

- GET /posts/{postId}/comments – Fetch comments
- POST /posts/{postId}/comments – Add a comment

These APIs follow REST principles using HTTP methods and JSON-based communication.

7. Database Design

The application uses a relational data model with the following entities:

Post Entity

- id
- content
- authorHandle
- createdAt
- likeCount
- commentCount

Comment Entity

- id
- content
- authorHandle
- createdAt
- post (Many-to-One relationship)

The relationship ensures each comment is associated with exactly one post.

8. Service Layer Logic

The service layer handles all business logic such as:

- Creating posts
- Incrementing like counts
- Fetching posts with updated comment counts
- Adding comments linked to posts

This approach prevents logic duplication and keeps controllers lightweight.

9. Frontend Design

The frontend is intentionally minimal and user-friendly.

Main Pages

1. Login Page (Demo-based)
2. Home Feed Page
3. Post Composer Section
4. Comment Interaction Section

The frontend interacts with the backend using JavaScript fetch() calls and dynamically updates the UI.

10. User Workflow

1. User enters a handle and logs in
2. User is redirected to the home feed
3. User creates a post
4. Other users can view and comment
5. Likes and comments update dynamically

This flow ensures simplicity and usability.

11. Testing and Validation

Testing was performed manually using:

- PowerShell commands (Invoke-WebRequest)
- Browser-based testing
- API response verification

Example:

GET http://localhost:8080/posts
Status Code: 200

All endpoints were validated for correct behavior and data persistence.

12. Security Considerations

The application intentionally avoids full authentication to maintain anonymity. However:

- Input validation is applied
- Database integrity is maintained
- Controlled API exposure is used

This approach balances simplicity and safety for academic use.

13. Performance Considerations

- Lightweight REST calls
- Efficient JPA queries
- File-based database for fast local access

The system performs well for small to medium datasets.

14. Limitations

The system lacks real authentication, moderation tools, and real-time features. The H2 database limits scalability. Content deletion is not supported. Automated testing is not implemented. These limitations are intentional to focus on core learning outcomes.

15. Future Enhancements

Future work includes JWT authentication, moderation tools, post deletion, WebSocket-based real-time updates, cloud deployment, mobile application support, and advanced analytics.

16. Conclusion

Unsaid successfully demonstrates how technology can be used to address emotional and social challenges while maintaining strong software engineering principles. The project integrates backend, database, and frontend components into a cohesive system. It serves as both a technical achievement and a socially meaningful application, making it a strong academic project.

17. References

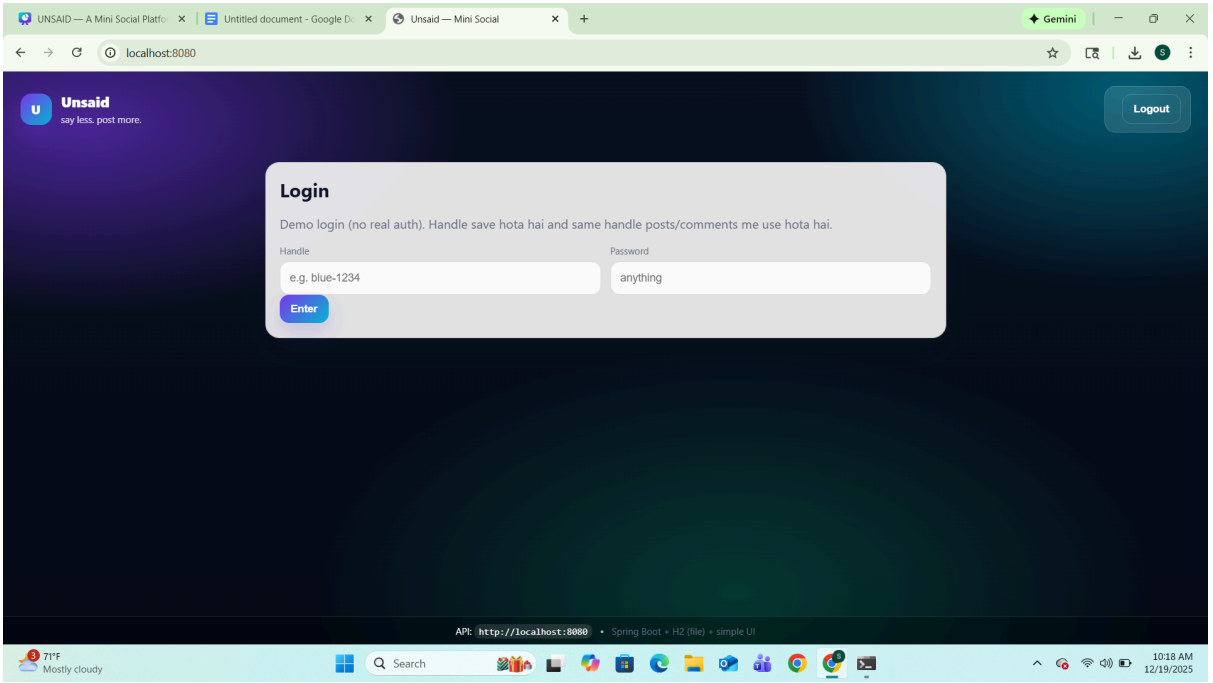
- Spring Boot Documentation
- Hibernate ORM Documentation
- REST API Design Principles
- H2 Database Documentation

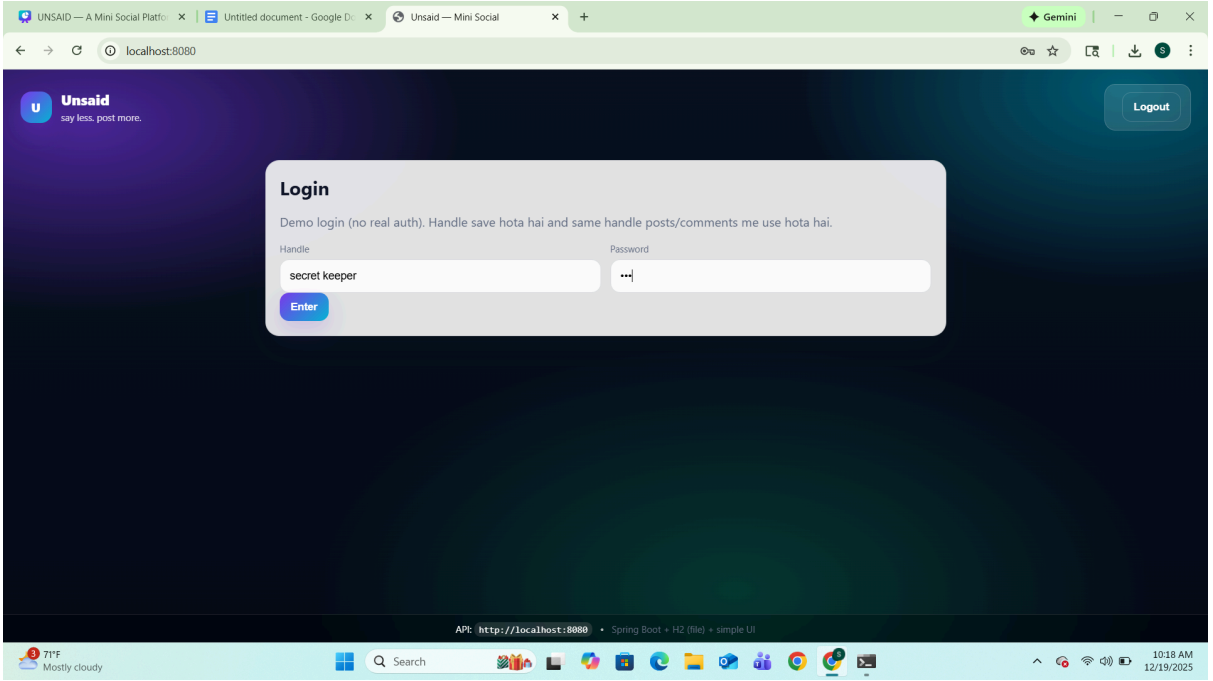
Screenshots:-

cd C:\Users\anush\Desktop\UnsaidApp

.\mvnw.cmd spring-boot:run

http://localhost:8080

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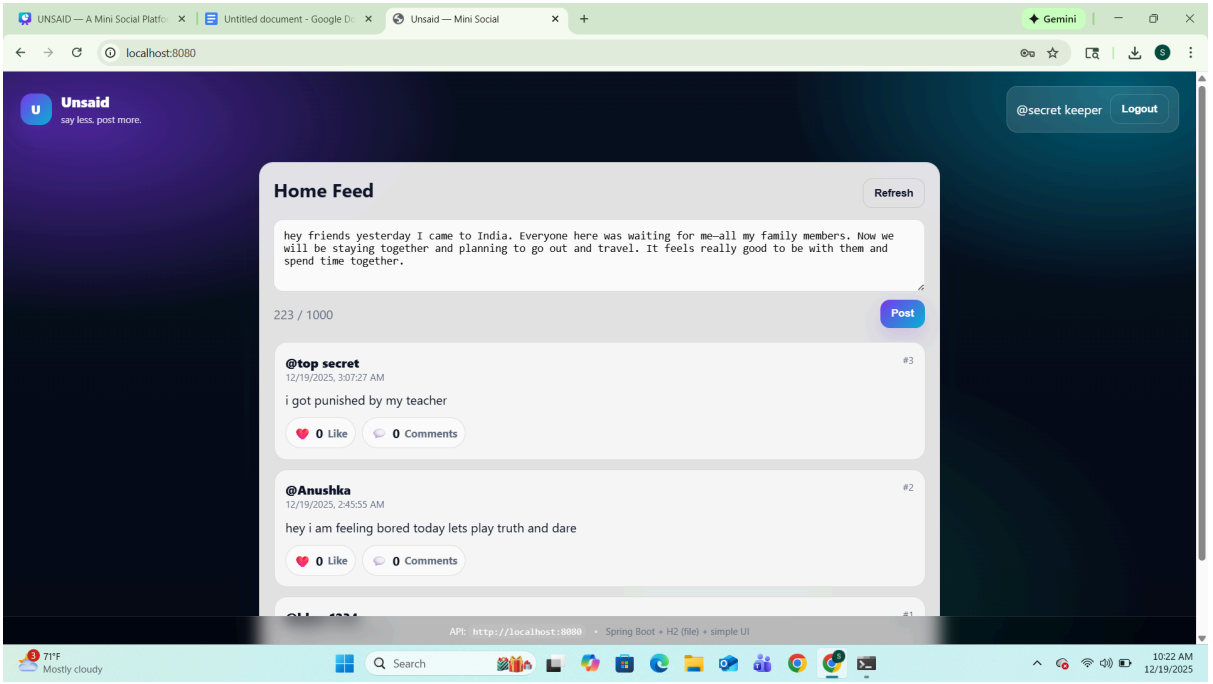
localhost:8080
Unsaid
say less, post more.
Logout
Login
Demo login (no real auth). Handle save hota hai and same handle posts/comments me use hota hai.
Handle
Password
secret keeper
Enter
API: http://localhost:8080 • Spring Boot + H2 (file) + simple UI
71°F
Mostly cloudy
Search
10:18 AM
12/19/2025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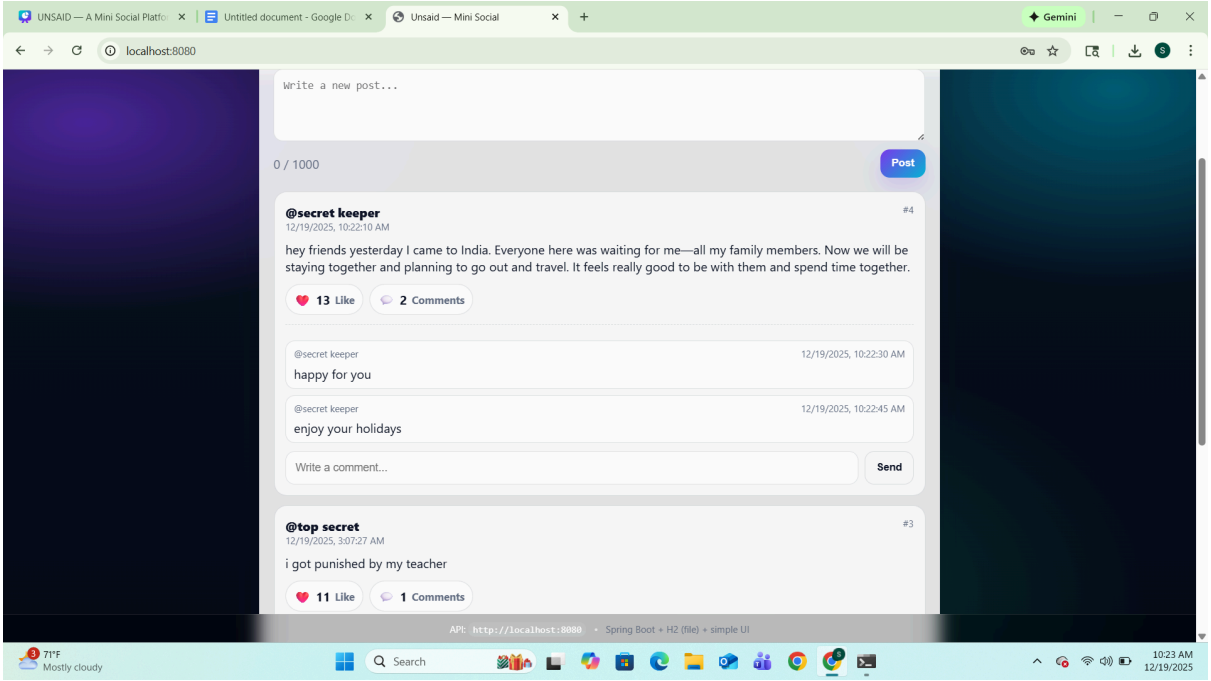
Gemini
localhost:8080
Unsaid
say less, post more.
@secret keeper
Logout
Home Feed
Refresh
hey friends yesterday I came to India. Everyone here was waiting for me—all my family members. Now we will be staying together and planning to go out and travel. It feels really good to be with them and spend time together.
223 / 1000
Post
@top secret
12/19/2025, 3:07:27 AM
#3
i got punished by my teacher
0 Like
0 Comments
@Anushka
12/19/2025, 2:45:55 AM
#2
hey i am feeling bored today lets play truth and dare
0 Like
0 Comments
API: http://localhost:8080 • Spring Boot + H2 (file) + simple UI
71°F
Mostly cloudy
Search
10:22 AM
12/19/2025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Gemini
localhost:8080
Write a new post...
0 / 1000
Post
@secret keeper
12/19/2025, 10:22:10 AM
#4
hey friends yesterday I came to India. Everyone here was waiting for me—all my family members. Now we will be staying together and planning to go out and travel. It feels really good to be with them and spend time together.
13 Like
2 Comments
@secret keeper
12/19/2025, 10:22:30 AM
happy for you
@secret keeper
12/19/2025, 10:22:45 AM
enjoy your holidays
Write a comment...
Send
@top secret
12/19/2025, 3:07:27 AM
#3
i got punished by my teacher
11 Like
1 Comments
API: http://localhost:8080 • Spring Boot + H2 (file) + simple UI
71°F
Mostly cloudy
Search
10:23 AM
12/19/2025